

LUCA MULTAZZU

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PROFESSIONAL EXPERIENCE

Backend Engineer & Applied Researcher — SICPA, Switzerland

September 2023 – Present

- Design and develop production-grade distributed systems using **Java 21, Spring Boot, Kafka, gRPC, REST, PostgreSQL**, and **AWS**.
- Contribute to architecture and development of high-availability systems across **payments, digital identity, and track-and-trace**.
- Conduct applied research at the intersection of **cryptography, digital identity, privacy-preserving technologies, and data science**.
- Research and prototype emerging technologies including **Zero-Knowledge Proofs, Verifiable Credentials, Digital Product Passports, privacy-preserving supply chains, offline payments, and CBDC systems**.
- Apply **Data Science and Machine Learning** techniques to security and fraud-detection problems.
- Contribute to **applied cryptography and privacy research**, translating emerging techniques into practical system concepts and prototypes.
- Work across **Java, Rust, Python, and Kotlin** in multidisciplinary research and engineering environments.

EDUCATION

MSc in Cyber Security — ETH Zurich (2021 – 2023)

Minor in Data Management Systems / GPA: 5.49 / 6

Key coursework: Applied Cryptography, Advanced Encryption Schemes, Big Data, Security Engineering, Distributed Systems, Formal Verification.

Joint degree with EPFL in Lausanne.

Selected Academic Projects:

- Optimal Distributed Read Atomic Transactions: Developed and tested a **novel read atomic algorithm** for distributed transactions.
- Road Segmentation Classifier: Built a neural network classifier for satellite imagery analysis.

TECHNICAL SKILLS

Programming:

Java, Kotlin, Rust, Python, C, C++, Scala

Backend & Distributed Systems:

Spring Boot 3, Kafka, gRPC, REST, Microservices, Event-Driven Architecture

Cryptography & Privacy:

Applied Cryptography, Zero-Knowledge Proofs, Digital Identity, Verifiable Credentials, Privacy-Preserving Technologies, Differential Privacy

Data & Machine Learning:

Data Science, Machine Learning, OSINT, Fraud Detection

Security & Identity:

Keycloak, OAuth 2.0, OpenID Connect, Authentication & Authorization, Secure Systems

Databases:

PostgreSQL, Oracle, Neo4j, GraphQL

Cloud & Infrastructure:

AWS, On-Premises, CI/CD, Jenkins

CERTIFICATIONS

Applied Cryptography

Specialisation — University of Colorado (2025)